Deliverable 7: 14-Component Rubric Matrix & Viva Voce Guide

Project: Intelligence Fusion Platform • **Course:** DBMS Mini Project (UE24CS395)
Authors: Saatvik Gupta (PES1UG24CS395) & Satyam Goyal (PES1UG24CS424) • PES University (2026–2027)

1. Complete 14-Component Evaluation Rubric Matrix (50 Marks)

#	Evaluation Criterion	Marks	Evidence & Implementation Files	Verification
1	ER Diagram	2	docs/er-diagram.md, 11 entities, 11 junctions	Verified Complete
2	Relational Schema & Keys	3	database/sql/schema.sql, strict PK/FK/CHECKs	Verified Complete
3	Data Integrity & Normalization	4	docs/database-design.md, 3NF/BCNF proofs	Verified Complete
4	Relational Implementation	5	PostgreSQL DDL & SQLAlchemy 22 models	Verified Complete
5	DML Operations	5	database/sql/dml-examples.sql (CRUD, SAVEPOINTs)	Verified Complete
6	Advanced SQL (SP, Function, Triggers, CTE)	6	database/sql/advanced-concepts.sql	Verified Complete
7	NoSQL Implementation	5	database/mongodb/ (schema, indexes, rud)	Verified Complete
8	Technology Justification	3	docs/technology-justification.md (ACID vs Flexible)	Verified Complete
9	Authorization & RBAC	3	database/sql/roles-and-grants.sql, JWT auth	Verified Complete
10	Next.js Forensic User Interface	5	frontend/ Next.js App Router, 16 views	Verified Complete
11	System Architecture & Integration	3	FastAPI gateway, Docker Compose setup	Verified Complete
12	Project Documentation & Repo Quality	3	README.md, 8 submission PDFs, clean Git	Verified Complete
13	Viva & Demonstration Preparation	3	docs/viva-preparation.md (30 Q&As;, 5-min walk)	Verified Complete
14	Team Contributions	Mandatory	docs/team-contributions.md (Saatvik & Satyam)	Verified Complete

2. Top 5 Viva Voce Questions & Definitive Examiner Answers

Q1: Why did you choose a Polyglot Persistence architecture instead of putting everything into PostgreSQL?

Relational SQL guarantees strict ACID constraints, foreign key cascades, and mathematical normalization required for evidentiary custody and legal admissibility. Conversely, MongoDB handles heterogeneous surveillance telemetry (CCTV logs, audio transcripts, sensor dumps) whose schema changes constantly, without requiring disruptive DDL migrations.

Q2: How does your schema achieve Boyce-Codd Normal Form (BCNF)?

Every relation is in 3NF and every determinant is a superkey. Many-to-many associations (such as suspects linked to multiple cases and vehicles) are isolated into 11 junction bridge tables with composite primary keys, eliminating partial and transitive functional dependencies.

Q3: How does the Recursive CTE function in your investigative link analysis?

It uses an anchor query selecting the initial suspect, then recursively executes union joins across the person_organizations and case_people bridge tables to discover multi-hop criminal associates up to 5 degrees of depth, terminating when no new associates are discovered.

Q4: How is evidence chain of custody cryptographically verified?

Each physical or digital evidence record stores an immutable SHA-256 hash calculated at collection time. Every transfer, checkout, or lab analysis event records an audit entry in evidence_custody with timestamp and responsible officer ID, ensuring complete tamper-evidence.

Q5: What occurs if PostgreSQL or MongoDB is offline during evaluation?

The platform implements zero-friction fault tolerance: if PostgreSQL is unreachable on port 5432, SQLAlchemy seamlessly falls back to a verified local SQLite database (intelligence_fusion.db) with identical schema, indexes, and constraints. If MongoDB is offline, the API cleanly reports standby status while all relational modules operate unimpeded.